

Jobs and Well-Being Measurement*

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Abstract

We measure well-being in a model in which individuals take one job in their ability set and have heterogenous preferences over jobs and how much they are paid.

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1 Introduction

2 The model

- Fixed universal set of existing jobs: $\mathcal{J}$.
- Pre-tax income function: $\mathbf{y} : \mathcal{J} \rightarrow \mathbb{R}_+$: $\mathbf{y}(j)$, also written $\mathbf{y}_j$, stands for pre-tax income associated to job j .
- Consumption set: $Z = \mathbb{R}_+ \times \mathcal{J}$, typical element $z = (c, j)$, where c is the consumption level and j the job to which the individual is assigned.
- Preferences: $R \in \mathcal{R}$, monotonic in the first argument.

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- Ability set $A \subseteq \mathcal{J}$.
- Well-being measure: $W : Z \times \mathcal{R} \times 2^{\mathcal{J}} \rightarrow \mathbb{R}$: $W(z, R, A)$ is the well-being level of an individual consuming bundle z , having preferences R and ability set A .
- Well-being is allowed to depend on how much jobs are paid $\mathbf{y}$, so we write $W(z, R, A; \mathbf{y})$.
- To save on notation, we assume that as soon as we measure $W(z, R, A; \mathbf{y})$, $z = (c, j)$ for some $j \in A$.

Remarks on the model:

- By allowing $\mathbf{y}$ to vary, we distinguish between non-pecuniary characteristics of jobs, assumed to be fixed, and pecuniary characteristics, which may vary (across regions or through time). Of course, with a sufficiently large $\mathcal{J}$ we can represent the universe of jobs by fixing $\mathbf{y}$, so that several jobs with exactly the same tasks to perform can be distinguished depending on how much they are paid. We prefer the latter definition, though, to be allowed to make pre-tax income vary without affecting the preferences of individuals towards a particular job.

3 Axioms

Basic axioms. First, W is a utility representation.

Representation: For all $z, z' \in Z$, $R \in \mathcal{R}$, $A \subseteq \mathcal{J}$, $\mathbf{y} \in \mathbb{R}_+^{\mathcal{J}}$,

$$z R z' \Leftrightarrow W(z, R, A; \mathbf{y}) \geq W(z', R, A; \mathbf{y}).$$

Job Neutrality: For all $z, z' \in Z$, $R \in \mathcal{R}$, $A, A' \subseteq \mathcal{J}$, $\mathbf{y} \in \mathbb{R}_+^{\mathcal{J}}$, if $z = (c, j)$, $z' = (c, j')$, $j' \notin A$, $A' = A \cup \{j'\}$, $\mathbf{y}(j) = \mathbf{y}(j')$ and $z I z'$, then

$$W(z, R, A; \mathbf{y}) = W(z', R, A'; \mathbf{y}).$$

I guess we will need axioms of Continuity.

We begin with compensation axioms.

Full Compensation: For all $z, z' \in Z$, $R \in \mathcal{R}$, $A, A' \subseteq \mathcal{J}$, $\mathbf{y}, \mathbf{y}' \in \mathbb{R}_+^{\mathcal{J}}$,

$$z I z' \Rightarrow W(z, R, A; \mathbf{y}) = W(z', R, A'; \mathbf{y}').$$

Independence of $\mathbf{y}$: For all $z \in Z$, $R \in \mathcal{R}$, $A \subseteq \mathcal{J}$, $\mathbf{y}, \mathbf{y}' \in \mathbb{R}_+^{\mathcal{J}}$,

$$W(z, R, A; \mathbf{y}) = W(z, R, A; \mathbf{y}').$$

Compensation for Reference Preferences.

Job j is irrelevant for an individual of type (R, A) at pre-tax incomes $\mathbf{y}(j)$ if $j \in A$ but this individual will never take job j : there is another job, j' , which this individual can take, which pays less, and independently of the tax (by tax progressivity the tax on j cannot be lower than that on j') this individual always prefers taking job j' : j is irrelevant for individual (R, A) , if $j \in A$ and there exists $j' \in A$ such that $\mathbf{y}(j') < \mathbf{y}(j)$ and for all $t \in \mathbb{R}$, $z, z' \in Z$ such that $z = (\mathbf{y}(j) - t, j)$ and $z' = (\mathbf{y}(j') - t, j')$: $z' P z$. For (R, A) we let $Irr(A; R, \mathbf{y})$ denote the set of irrelevant jobs for an individual of type (R, A) facing pre-tax incomes $\mathbf{y}$.

Independence of Irrelevant Jobs: For all $z = (c, j) \in Z$, $R \in \mathcal{R}$, $A \subseteq \mathcal{J}$, $j' \in A$, $\mathbf{y} \in \mathbb{R}_+^{\mathcal{J}}$, if $j \neq j'$ and $j' \in Irr(A; R, \mathbf{y})$, then

$$W(z, R, A; \mathbf{y}) = W(z, R, A \setminus \{j'\}; \mathbf{y})$$

Full Compensation implies Independence of $\mathbf{y}$, Compensation for Reference Preferences and Independence of Irrelevant Jobs.

Next, responsibility axioms. We use the following terminology: $\max_R\{A; \mathbf{y}\}$ stands for any of the preferred bundles among those composed of jobs in A and their pre-tax income in $\mathbf{y}$, absent any taxation. Note that if there are several bundles that maximize R in A , they are indifferent to each other.

Full Responsibility: For all $R, R' \in \mathcal{R}$, $A \subseteq \mathcal{J}$, $\mathbf{y} \in \mathbb{R}_+^{\mathcal{J}}$,

$$W(\max_R\{A; \mathbf{y}\}, R, A; \mathbf{y}) = W(\max_{R'}\{A; \mathbf{y}\}, R', A; \mathbf{y}).$$

Responsibility For Equal Pay: For all $R, R' \in \mathcal{R}$, $A \subseteq \mathcal{J}$, $\mathbf{y} \in \mathbb{R}_+^{\mathcal{J}}$ such that $\mathbf{y}(j) = \mathbf{y}(j')$ for all $j, j' \in A$,

$$W(\max_R\{A; \mathbf{y}\}, R, A; \mathbf{y}) = W(\max_{R'}\{A; \mathbf{y}\}, R', A; \mathbf{y}).$$

Responsibility for acquired ability (RAA) We model ability sets as partly the outcome of past choices and partly due to circumstances, we introduce a reduced-

form *ability-formation correspondence*. Let Θ denote the set of circumstances (non-responsibility factors such as health shocks, family background constraints, and labour-market frictions), and let E denote the set of merit/investment levels (education, training, and job experience). The observed ability set is generated by a correspondence

$$A : \Theta \times E \rightarrow 2^{\mathcal{J}}, \quad (\theta, e) \mapsto A(\theta, e) \subseteq \mathcal{J}. \quad (1)$$

We impose the following responsibility requirement: for any preferences $R \in \mathcal{R}$, any income function $\mathbf{y} \in \mathbb{R}_+^{\mathcal{J}}$, any circumstances $\theta \in \Theta$, and any two feasible merit levels $e, e' \in E(\theta)$,

$$W(\max_R\{A(\theta, e); \mathbf{y}\}, R, A(\theta, e); \mathbf{y}) = W(\max_R\{A(\theta, e'); \mathbf{y}\}, R, A(\theta, e'); \mathbf{y}), \quad (2)$$

that is, conditional on the same circumstances θ , differences in the achieved ability set that arise from different feasible merit choices should not affect the well-being level assigned at the laissez-faire.

4 Measures

Measure 1: We restrict our attention to the individual's ability set, without looking at the pre-tax incomes associated to these jobs, and we look at the preferred job and its corresponding consumption level that leaves the individual indifferent to their current bundle.

$$W(z, R, A; \mathbf{y}) = w \Leftrightarrow z I \max_R\{B\}$$

where

$$z' = (c', j') \in B \Leftrightarrow j' \in A \text{ and } c' = w.$$

If I'm right, this measure satisfies Independence of $\mathbf{y}$ and Responsibility For Equal Pay. I am confident it is the only measure to satisfy these two axioms. To be proven.

Measure 2: We restrict our attention to the individual's ability set, we look at the preferred job in this set to which the individual is indifferent to their current bundle, where "preferred job" means that the tax (or subsidy) that would leave them indifferent is minimal, and we apply this tax (or subsidy) to all the jobs in the ability set to identify the "best-paid equivalent" job, the pay of which is the well-being of

the individual.

$$W(z, R, A; \mathbf{y}) = w \Leftrightarrow z \underset{R}{I} \max\{B\}$$

where there exists $t \in \mathbb{R}$ such that

$$z' = (c', j') \in B \Leftrightarrow j' \in A \text{ and } c' = \mathbf{y} - t,$$

and

$$w = \max_{j \in A} \mathbf{y}(j) - t.$$

If I'm right, this measure satisfies Full Responsibility and Compensation for Reference Preferences if the reference preferences are $R^c \in \mathcal{R}$ defined by

$$z = (c; j) R^c z' = (c', j') \Leftrightarrow c \geq c'.$$

I am reasonably confident it is the only measure to satisfy these two axioms and some other ones that will extend Job Neutrality. To be proven.

Measure 3: The Laisser-Faire measure. All individuals have the same well-being at laissez-faire. Otherwise, compute the amount of tax t that leaves the individual indifferent between their current bundle and maximizing over their ability set and maying t (better to write it as a subsidy I guess).

$$W(z, R, A; \mathbf{y}) = w \Leftrightarrow z \underset{R}{I} \max\{B\}$$

where

$$z' = (c', j') \in B \Leftrightarrow j' \in A \text{ and } c' = \mathbf{y}(j') + w.$$

I think the Laisser-Faire measure satisfies Full Responsibility and Independence of Irrelevant Jobs and it is the only one that satisfies these two axioms